Laborbericht

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Praktikum	Hochfrequenzschaltungstechnik	Experiment Nr.	5
Matr. Nr.	35690104	Datum	2.2.2026

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Jeglicher im Bericht verwendeter, von KI erzeugter Text ist als direktes Zitat zu betrachten und muss daher durch Anführungszeichen und eine entsprechende Quellenangabe kenntlich gemacht werden. Die Quellenangabe muss Autor, Prompt, Herausgeber, Datum und die URL des verwendeten Tools enthalten. Ein Beispiel zur Zitierung von KI-generiertem Text finden Sie im Folgenden:

1. ChatGPT, Antwort auf "Erklär mir, wie man aus Haushaltszutaten einen Pizzateig zubereitet," OpenAI, März 7, 2023, https://chat.openai.com/chat.

Datum, Unterschrift

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Praktikum Hochfrequenz-Schaltungstechnik WS25/26

Bericht

Name:	Sebastian Grigorevski	**Matrikelnr.:**	35690104
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5.5.1

According to the approximation formula of the rise time of DSOs

$$t_o = \frac{0.35}{BW}$$

the rise time for a 60 MHz (120 MHz) DSO are $t_o = 5.8\,\mathrm{ns}$ ($t_o = 2.9\,\mathrm{ns}$) [1]. Thus the DSO cannot resolve rise times lower than those calculated values. The rise and fall times of the inverter are much faster at 2.05 ns and 1.90 ns (measured with a 350 MHz DSO). Since the rise and fall times are significantly faster the given DSOs are not able to measure them correctly. This can be confirmed by a measurement with a 60 MHz where the rise time was noted at 5.5 ns and the fall time was 4.5 ns being much higher than the actual value. On the other hand the length of the pulse was measured at 10 ns with both DSOs. This suggests that the rise time of the DSO does not influence this measurement.

5.5.2

The formula for reflection coefficients is stated as

$$\Gamma = \frac{Z_\mathrm{t} - Z_0}{Z_\mathrm{t} + Z_0}$$

where Z_t is the termination impedance and $Z_0 = 50\,\Omega$ is the impedance of the wire [1]. For different termination impedances the theoretical reflection coefficients $\Gamma_{\mathrm{theo.}}$ are calculated in table 1. Further the measured voltage peaks of the signal and the reflection were used to determine the measured reflection coefficients in the table by dividing their values. This was done for the same terminations

Table 1: theoretical and measured reflection coefficients at a wire impedance of $Z_0 = 50\,\Omega$

Z_t	$50\,\Omega$	$75\,\Omega$	$93\,\Omega$
$\Gamma_{\mathrm{theo.}}$	0	0.20	0.30
$\Gamma_{\mathrm{measured}}$	0	0.12	0.19

Differences between the theoretical and measured values can occur due to parasitic losses in the wire since the dielectric material between inner wire and cylindrical conductor deafens the signal due to its magnetic properties.

5.5.3

- Measure the time difference Δt between the incident and reflected pulse on the oscilloscope.
- Convert the oscilloscope reading from divisions to time:

$$\Delta t = N_{\text{Div}} \cdot t_{\text{Div}}$$

- Take measurements for both open and short-circuited cable ends.
- Calculate the mean value to reduce measurement uncertainty:

$$\Delta t_{\text{mean}} = \frac{\Delta t_{\text{open}} + \Delta t_{\text{short}}}{2}$$

- Determine the cable length using the cable's velocity factor k and :

$$l = \frac{c \cdot k \cdot \Delta t}{2},$$

  where c is the speed of light and k is the velocity factor of the cable and the pulse travels through the cable two times (reflection).

Cable type	Δt [ns]	k	calc. Length l [m]	Measured Length l [m]
RG58	38.5	0.66	3.8	3.97
RG188	42.5	0.70	4.4	4.52
SAT (white)	30.5	0.79	–	3.59

Table 2: Cable lengths calculated from propagation delay measurements

The results obtained for open and short-circuited cable terminations agree within the measurement uncertainty. Remaining deviations are mainly caused by reading errors, additional cable lengths due to T-connectors (and ikea paper measuring tapes :).

5.5.4

Same formula from previous section is used to determine the length of the cable drum. Below is a sketch of the measurement setup. The determined length is

$$l = \frac{c \cdot 0.66 \cdot 405\,\text{ns}}{2} = 40,07\,\text{m}.$$

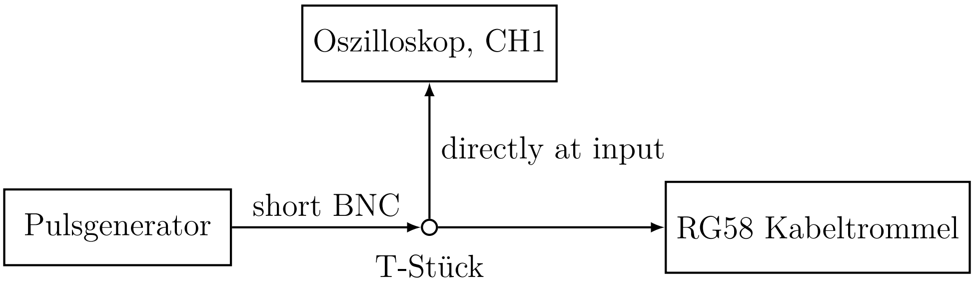

Figure 1: Measurement setup of length measurement

5.5.5

The time needed for a signal to reach the end of a 5 m RG-58 cable is $t = \frac{5m}{0.66 \cdot c} \approx 25$ ns thus the length of the pulse is 250 ns.

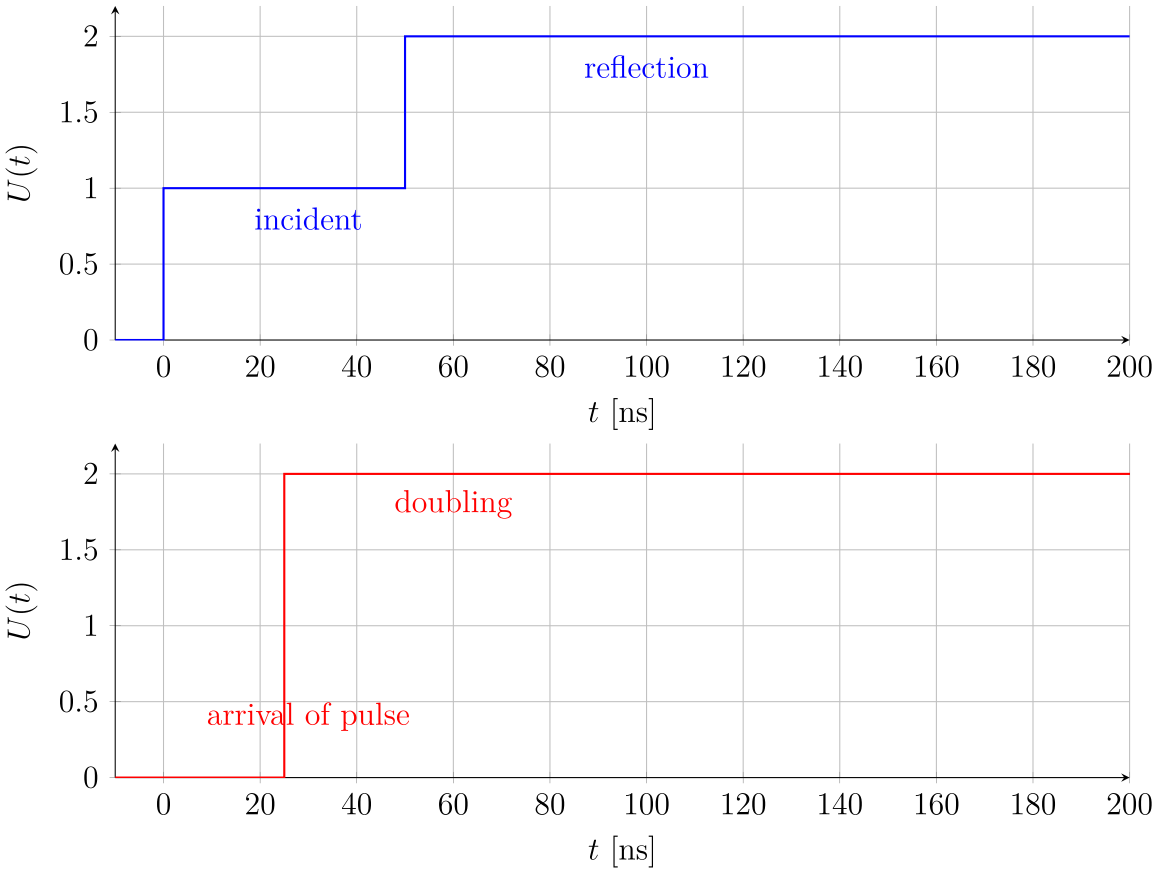

Figure 2: Top: signal at generator. Bottom: signal at the end of a 5 m RG-58 cable.

5.5.6

Three different termination impedances 50, 75 and 93 Ω were connected to the SAT cable to narrow down the impedance range of the cable. Due to the reflections we know that it has to be between 50 and 75 Ω. Using the potentiometer we were able to find the configuration where complete termination occurs and determined the impedance of the SAT cable at 72.8 $\pm 0.2\,\Omega$.

5.5.7

The velocity factor k of the SAT cable can be determined from the known physical length s and the measured round-trip time difference Δt between the incident and reflected pulse. Rearranging the propagation relation

$$v = k\,c = \frac{2s}{\Delta t}$$

gives

$$k = \frac{v}{c} = \frac{2s}{c\,\Delta t}.$$

Inserting the values from table 2, $s = 3.59\,\mathrm{m}$ and $\Delta t = 30.5\,\mathrm{ns}$, yields

$$k = \frac{2 \cdot 3.59\,\mathrm{m}}{(3.00 \times 10^{8}\,\mathrm{m/s}) \cdot 30.5 \times 10^{-9}\,\mathrm{s}} \approx 0.79.$$

5.5.8

Assuming that the dielectric of the SAT cable consists of polyethylene, the velocity factor k of an electromagnetic wave in the cable is given by

$$k = \frac{v}{c} = \frac{1}{\sqrt{\varepsilon_r \mu_r}}\,,$$

where c is the speed of light in vacuum, ε_r is the relative permittivity of the dielectric and μ_r its relative permeability [4]. For non-magnetic dielectrics such as polyethylene we can set $\mu_r \approx 1$, so that

$$k \approx \frac{1}{\sqrt{\varepsilon_r}}\,.$$

Using a typical value of $\varepsilon_r \approx 2.3$ for *solid* polyethylene [2], the theoretical velocity factor becomes

$$k_{\text{solid PE}} = \frac{1}{\sqrt{2.3}} \approx 0.66\,.$$

This is the value expected for a coaxial cable with a solid polyethylene dielectric.

From our on the SAT cable (white), we obtained a significantly higher velocity factor of about

$$k_{\text{meas}} \approx 0.79\,.$$

The corresponding effective relative permittivity of the dielectric is

$$\varepsilon_{r,\text{eff}} = \frac{1}{k_{\text{meas}}^2} \approx \frac{1}{0.79^2} \approx 1.6\,.$$

This effective permittivity is much lower than the value for solid polyethylene. Comparing the theoretical velocity factor for solid polyethylene with the measured value, we conclude that the dielectric of the SAT cable is not solid but foamed polyethylene (with air).

5.5.9

The capacitance per unit length C', inductance per unit length L', and characteristic impedance Z_0 of a coaxial cable are given by

$$L' = \frac{\mu}{2\pi}\ln\left(\frac{r_2}{r_1}\right), C' = \frac{2\pi\varepsilon}{\ln\left(\frac{r_2}{r_1}\right)}, Z_0 = \sqrt{\frac{L'}{C'}}[3] \tag{1}$$

Substituting these equations into the impedance yields

$$Z_0 = \frac{1}{2\pi}\ln\left(\frac{r_2}{r_1}\right)\sqrt{\frac{\mu_r\mu_0}{\varepsilon_r\varepsilon_0}} \tag{2}$$

Solving for the relative permittivity and inserting the numerical values gives

$$\varepsilon_r = \left(\frac{1}{2\pi Z_0}\ln\left(\frac{r_2}{r_1}\right)\right)^2\frac{\mu_r\mu_0}{\varepsilon_0} \approx 1.36 = \left(\frac{1}{2\pi\cdot 75\,\Omega}\ln\left(\frac{4.3\,\mathrm{mm}}{1\,\mathrm{mm}}\right)\right)^2\frac{\mu_0}{\varepsilon_0} \approx 1.36 \tag{3}$$

5.5.10

For a lossless coaxial cable the characteristic impedance Z_0 is related to the per-unit-length inductance L' and capacitance C' by

$$Z_0 = \sqrt{\frac{L'}{C'}}\,.$$

Solving for L' gives

$$L' = Z_0^2\,C'\,.$$

Using the SAT cable impedance $Z_0 \approx 75\,\Omega$ and the given capacitance per unit length $C' = 56\,\mathrm{pF\,m^{-1}} = 56\cdot 10^{-12}\,\mathrm{F\,m^{-1}}$, we obtain

$$L' = 0.315\,\mathrm{\mu H\,m^{-1}}.$$

5.5.11

As shown in the figure below, the repetition rate of V_{Caps} is significantly faster when the 1N4148M diode is used, whereas the 1N6263 maintains a comparatively lower repetition rate. This behavior is due to differences in transit time between the silicon diode and the Schottky diode. The transit time is the time required for charge carriers to traverse the depletion region of the diode and for stored charge to be removed during switching, which directly limits the maximum achievable switching speed. The 1N4148M diode has a longer transit time allowing the capacitors to charge up longer.

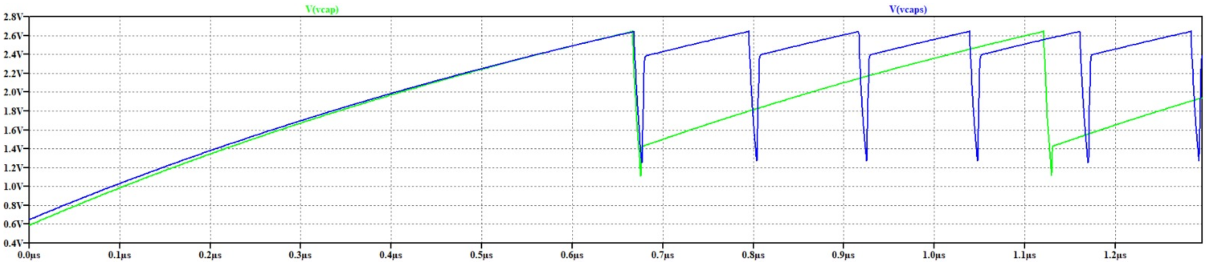

Figure 3: Simulation of the pulse generator (V(cap) using the 1N6263 diode, V(caps) using the 1N4148M diode).

References

[1] Axel Bangert. *Praktikum Hochfrequenz-Schaltungstechnik*. Kassel.

[2] Editor Engineeringtoolbox. Relative permittivity - the dielectric constant. 9/9/2010.

[3] Andreas Thiede. *Integrierte Hochfrequenzschaltkreise*. Springer Fachmedien Wiesbaden, Wiesbaden, 2013.

[4] Wikipedia. Verkürzungsfaktor — Wikipedia, the free encyclopedia. `http://de.wikipedia.org/w/index.php?title=Verk%C3%BCrzungsfaktor&oldid=259972853`, 2026. [Online; accessed 28-January-2026].